

4-5

Solving Rational Equations

I CAN... solve rational equations and identify extraneous solutions.

VOCABULARY

- rational equation
- extraneous solution

CRITIQUE & EXPLAIN

Nicky and Tavon used different methods to solve the equation $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$.

Nicky	Tavon
$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$	$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$
$\frac{1}{2}x = \frac{9}{10} - \frac{2}{5}$	$10\left(\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}\right)$
$\frac{1}{2}x = \frac{5}{10}$	$5x + 4 = 9$
$x = 1$	$5x = 5$
	$x = 1$
The solution is 1.	The solution is 1.

- Explain the different strategies that Nicky and Tavon used and the advantages or disadvantages of each.
- Did Nicky use a correct method to solve the equation? Did Tavon?
- Use Structure** Why might Tavon have chosen to multiply both sides of the equation by 10? Could he have used another number? Explain.

ESSENTIAL QUESTION

How can you solve rational equations and identify extraneous solutions?

EXAMPLE 1 Solve a Rational Equation

What is the solution to each rational equation?

A. $\frac{1}{x+4} = 2$

$$\begin{aligned}(x+4)\left(\frac{1}{x+4}\right) &= 2(x+4) \\ 1 &= 2x + 8 \\ x &= -\frac{7}{2}\end{aligned}$$

The solution is $x = -\frac{7}{2}$.

B. $\frac{1}{x-3} = 5$

$$\begin{aligned}(x-3)\left(\frac{1}{x-3}\right) &= 5(x-3) \\ 1 &= 5x - 15 \\ x &= \frac{16}{5}\end{aligned}$$

The solution is $x = \frac{16}{5}$.

Multiply both sides of the equation by the common denominator to eliminate the fractions. Then solve. Confirm that the solution is valid in the original equation.

VOCABULARY

A **rational equation** is an equation that contains a rational expression.

Try It! 1. What is the solution to each equation?

a. $\frac{2}{x+5} = 4$

b. $\frac{1}{x-7} = 2$

EXAMPLE 2 Solve a Work-Rate Problem

Arthur and Cheyenne are making Chinese paper cutting designs to give as gifts to wish family and friends good luck in the coming year. Together they can complete the job in 24 hours. How long would it take Cheyenne to complete the job if she was working alone?



Step 1 Determine the work-rates of Arthur and Cheyenne.

Let x represent the number of hours Arthur needs to finish the job himself.

Arthur works at a rate of $\frac{1 \text{ job}}{x \text{ hours}} = \frac{1}{x} \frac{\text{jobs}}{\text{hour}}$.

The amount of a job completed per hour is the work-rate.

Cheyenne is twice as fast, so she works at a rate of $\frac{1 \text{ job}}{\frac{1}{2}x \text{ hours}} = \frac{2}{x} \frac{\text{jobs}}{\text{hour}}$.

Working together, they complete the job in 24 hours, so the work-rate is $\frac{1 \text{ job}}{24 \text{ hours}} = \frac{1}{24} \frac{\text{job}}{\text{hour}}$.

Step 2 Write the equation for their rates working together.

$$\begin{aligned}\frac{1}{x} + \frac{2}{x} &= \frac{1}{24} \\ 24x\left(\frac{1}{x} + \frac{2}{x}\right) &= \left(\frac{1}{24}\right)24x \\ 24 + 48 &= x \\ 72 &= x\end{aligned}$$

Write the equation for their rates working together. Then solve.

It takes Arthur 72 hours to create the gifts alone. Since Cheyenne works twice as fast as Arthur, it would take her 36 hours to create the gifts alone.

COMMON ERROR

You might have multiplied x by 2 because Cheyenne works twice as fast. However, it takes Cheyenne *half* the time to make the gifts.

Try It! 2. It takes 12 hours to fill a pool with two pipes, where the water in one pipe flows three times as fast as the other pipe. How long will it take the slower pipe to fill the pool by itself?

EXAMPLE 3 Identify an Extraneous Solution

What is the solution of the equation $\frac{1}{x-5} + \frac{x}{x-3} = \frac{2}{x^2 - 8x + 15}$?

Step 1 Multiply each side of the equation by the common denominator, $(x-5)(x-3)$.

$$(x-5)(x-3)\left(\frac{1}{x-5} + \frac{x}{x-3}\right) = \frac{2(x-5)(x-3)}{x^2 - 8x + 15}$$

Step 2 Continue to simplify.

$$\frac{(x-5)(x-3)}{x-5} + \frac{x(x-5)(x-3)}{x-3} = \frac{2(x-5)(x-3)}{x^2 - 8x + 15}$$

Step 3 Divide out common factors in the numerator and the denominator.

$$\frac{\cancel{(x-5)}(x-3)}{\cancel{x-5}} + \frac{x\cancel{(x-5)}\cancel{(x-3)}}{\cancel{x-3}} = \frac{2\cancel{(x-5)}\cancel{(x-3)}}{\cancel{x^2 - 8x + 15}}$$

Step 4 Solve the equation.

$$(x-3) + x(x-5) = 2$$

$$x-3 + x^2 - 5x = 2$$

$$x^2 - 4x - 3 = 2$$

$$x^2 - 4x - 5 = 0$$

$$(x-5)(x+1) = 0$$

$$x = 5 \text{ and } x = -1$$

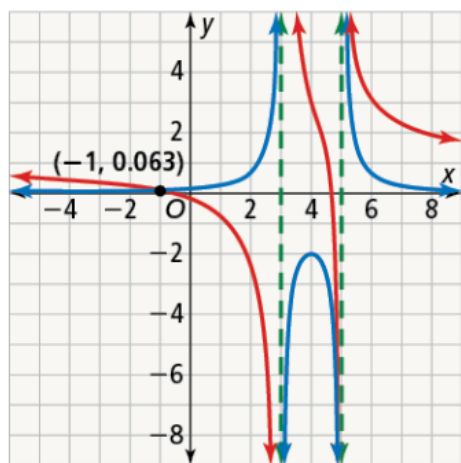
You can divide out common factors under the assumption that $\frac{(x-5)}{(x-5)} = 1$ and $\frac{(x-3)}{(x-3)} = 1$. This is only true if $x \neq 5$ or 3 .

Consider both solutions. If either solution makes the value of the denominator 0, it is not valid.

The solution $x = 5$ is an **extraneous solution** because it is not in the domain of the original equation.

The solution of the equation $\frac{1}{x-5} + \frac{x}{x-3} = \frac{2}{x^2 - 8x + 15}$ is -1 .

To see that there is only one solution, graph $f(x) = \frac{1}{x-5} + \frac{x}{x-3}$ and $g(x) = \frac{2}{x^2 - 8x + 15}$.



LOOK FOR RELATIONSHIPS

How are an extraneous solution and an asymptote related? Is this always true?

Both graphs have vertical asymptotes at $x = 3$ and $x = 5$. Therefore, neither graph has a value at $x = 5$. The graphs only intersect in one point, at $x = -1$.

Try It! 3. What is the solution to the equation $\frac{1}{x+2} + \frac{1}{x-2} = \frac{4}{(x+2)(x-2)}$?

EXAMPLE 4 Check for Extraneous Solutions**HAVE A GROWTH MINDSET**

In what ways do you give your best effort and persist?

What are the solutions to the following equations?

A. $\frac{5x}{x-2} = 7 + \frac{10}{x-2}$

$$\frac{5x}{x-2} = 7 + \frac{10}{x-2}$$

$$(x-2)\left(\frac{5x}{x-2}\right) = \left(7 + \frac{10}{x-2}\right)(x-2)$$

Remember to multiply all terms on both sides of the equation by the least common denominator.

$$5x = 7(x-2) + 10$$

Distribute, collect like terms, and solve for x .

$$5x = 7x - 14 + 10$$

$$-2x = -4$$

$$x = 2$$

Check the solution in the original equation. The value 2 is an extraneous solution because it would cause the denominator in the original equation to be equal to 0.

This equation has no solution.

B. $\frac{3}{x-3} = \frac{x}{x-3} - \frac{x}{4}$

$$\frac{3}{x-3} = \frac{x}{x-3} - \frac{x}{4}$$

$$(4)(x-3)\left(\frac{3}{x-3}\right) = \left(\frac{x}{x-3} - \frac{x}{4}\right)(4)(x-3)$$

Multiply by the LCD.

$$4(3) = 4(x) - x(x-3)$$

$$12 = 4x - x^2 + 3x$$

$$x^2 - 7x + 12 = 0$$

Write in standard form so that the quadratic is equal to 0. Then factor the polynomial and solve for x .

$$(x-3)(x-4) = 0$$

$$x-3=0 \text{ or } x-4=0$$

$$x=3 \text{ or } x=4$$

Check the solutions in the original equation. The value 3 is an extraneous solution because it would cause the denominator of the original equation to be equal to zero. The only solution to the equation is $x=4$.

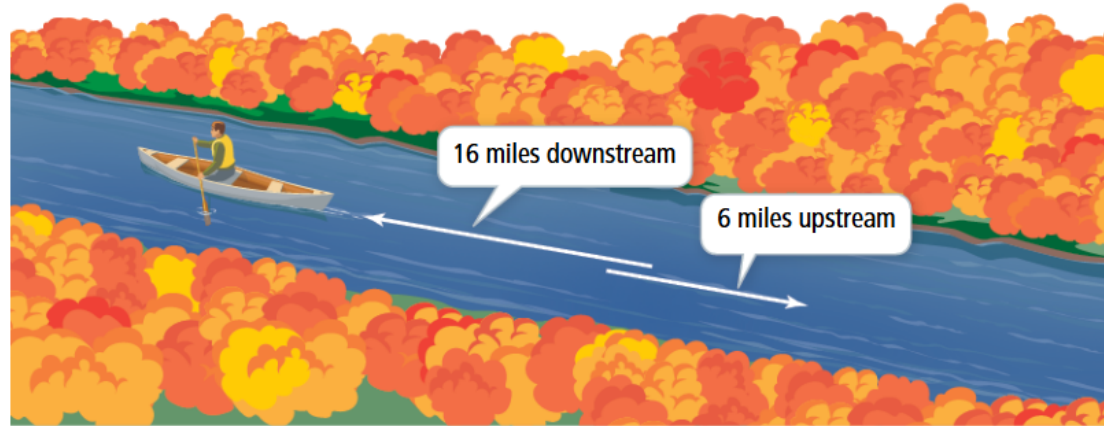
Try It! 4. What are the solutions to the following equations?

a. $x + \frac{6}{x-3} = \frac{2x}{x-3}$

b. $\frac{x^2}{x+5} = \frac{25}{x+5}$

EXAMPLE 5 Solve a Rate Problem

Paddling with the current in a river, Jake traveled 16 miles. Even though he paddled upstream for an hour longer than the amount of time he paddled downstream, Jake could only travel 6 miles against the current. In still water, Jake paddles at a rate of 5 mph. What is the speed of the current in the river?



Formulate

Let c be the rate of the river's current in miles per hour.

Recall that distance = rate $\cdot$ time, so time = $\frac{\text{distance}}{\text{rate}}$.

Jake's paddle rate in still water + the rate of the current

$$\frac{16}{5+c} + 1 = \frac{6}{5-c}$$

Jake paddles 1 h longer upstream than downstream.

Jake's paddle rate in still water – the rate of the current

Compute

Solve the equation for c .

$$\frac{16}{5+c} + 1 = \frac{6}{5-c}$$

$$(5+c)(5-c)\left(\frac{16}{5+c} + 1\right) = \left(\frac{6}{5-c}\right)(5+c)(5-c)$$

$$80 - 16c + 25 - c^2 = 30 + 6c$$

$$0 = c^2 + 22c - 75$$

$$0 = (c+25)(c-3)$$

$$0 = c+25 \quad 0 = c-3$$

$$c = -25 \quad c = 3$$

Interpret

The solution $c = -25$ is extraneous because the speed of the current cannot be negative.

The speed of the current is 3 mph.

- Try It!** 5. Three people are planting tomatoes in a community garden. Marta takes 50 minutes to plant the garden alone, Benito takes x minutes and Tyler takes $x + 15$ minutes. If the three of them take 20 minutes to finish the garden, how long would it have taken Tyler alone?

CONCEPT SUMMARY Solving Rational Equations

WORDS

A rational equation is an equation that contains a rational expression. To solve, identify the domain for the variable. Then multiply both sides of the equation by a common denominator and solve. An extraneous solution is a solution that is not valid because that value is excluded from the domain of the original equation.

ALGEBRA

$$\frac{1}{x} + \frac{2}{x} = \frac{1}{6}$$

Domain: $x \neq 0$

$$6x\left(\frac{1}{x} + \frac{2}{x}\right) = 6x\left(\frac{1}{6}\right)$$

$$6 + 12 = x$$

$$18 = x$$

The domain includes $x = 18$, so the solution to the equation is 18.

$$\frac{x^2 + 4}{x - 1} = \frac{5}{x - 1}$$

Domain: $x \neq 1$

$$(x - 1)\left(\frac{x^2 + 4}{x - 1}\right) = (x - 1)\left(\frac{5}{x - 1}\right)$$

$$x^2 + 4 = 5$$

$$x^2 = 1$$

$$x = \pm 1$$

The domain does not include $x = 1$, so 1 is an extraneous solution. It does include $x = -1$, so the solution to the equation is -1 .

Do You UNDERSTAND?

- ESSENTIAL QUESTION** How can you solve rational equations and identify extraneous solutions?
- Vocabulary** Write your own example of a rational equation that, when solved, has at least one **extraneous solution**.
- Error Analysis** A student solved the rational equation as follows:

$$\frac{1}{2x} - \frac{2}{5x} = \frac{1}{10x} - 3; x = 0$$

Describe and correct the error the student made. © MP3

- Construct Arguments** Yuki says, "You can check the solution(s) of rational equations in any of the steps of the solution process." Explain why her reasoning is incorrect. © MP3

Do You KNOW HOW?

Solve.

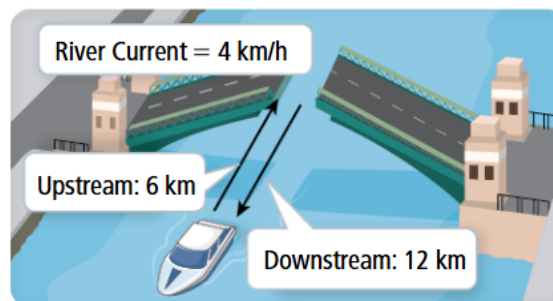
$$5. \frac{4}{x+6} = 2$$

$$6. \frac{x^2}{x+3} = \frac{9}{x+3}$$

- Organizing given information into a table can be helpful when solving rate problems. Use this table to solve the following problem.

	Distance	Rate	Time
Upstream			
Downstream			

A boat can travel 6 km upstream in the same time it takes to travel 12 km downstream. Find the speed of the boat in still water.



UNDERSTAND

8. **Reason** If you solve a work-rate problem and your solution, which represents the amount of time it would take *working together*, exceeds the individual *working alone* times that are given, then how do you know your solution is unreasonable? Explain.
9. **Construct Arguments** Explain why a negative solution must be eliminated as an extraneous solution when solving a rational equation for an unknown rate.
10. **Error Analysis** Describe and correct the error Miranda made in solving the rational equation.

$$\begin{aligned}\frac{1}{x-2} + \frac{x-2}{x+2} &= \frac{x-4}{x-2} \\ (x+2)(x-2)\left(\frac{1}{x-2} + \frac{x-2}{x+2}\right) &= \left(\frac{x-4}{x-2}\right)(x+2)(x-2) \\ (x+2)(1) + (x-2)(x-2) &= (x+4)(x+2) \\ x+2+x^2-4x+4 &= x^2+6x+8 \\ -3x+6 &= 6x+8 \\ -2 &= 9x; \text{ or } x = -\frac{2}{9}\end{aligned}$$

11. **Generalize** In addition to identifying extraneous solutions, why else is it important to substitute your solution into the original equation?
12. **Mathematical Connections** Explain how solving rational equations is related to solving linear and quadratic equations.
13. **Higher Order Thinking** Write a rational equation that cannot have 2 or -6 as solutions.
14. **Make Sense and Persevere** Solve the rational equation shown. Explain what is unique about the solution.

$$\frac{x^2 - 7x - 18}{x + 2} = x - 9$$

PRACTICE

Solve the equation. SEE EXAMPLE 1

15. $\frac{1}{x-3} = 10$

16. $\frac{15}{x+3} = 3$

17. $\frac{12}{x-4} = 9$

18. $\frac{5}{3-x} = 1$

Solve the problem. SEE EXAMPLE 2

19. Working individually, Paige and Malia can each complete a landscaping job in the time shown. Working together, how long would it take them to complete the job?



20. Russel and Aaron can build a shed in the time shown when working together. Aaron works three times as fast as Russel. How long would it take Russel to build the shed if he were to work alone?



Solve the equation. SEE EXAMPLE 3

21. $\frac{x}{x-3} - 4 = \frac{3}{x-3}$

22. $\frac{x^2}{x-10} = \frac{100}{x-10} - 10$

Solve the equation. SEE EXAMPLE 4

23. $\frac{4}{3(x+1)} = \frac{12}{x^2-1}$

24. $\frac{x}{x-3} + \frac{2x}{x+3} = \frac{18}{(x+3)(x-3)}$

Solve the problem. SEE EXAMPLE 5

25. A boat travels 8 miles upstream in the same amount of time it can travel 12 miles downstream. In still water the speed of the boat is 5 mi/h. What is the speed of the current?

PRACTICE & PROBLEM SOLVING

APPLY

25. **Make Sense and Persevere** Kenji can finish a puzzle in 2 hours working alone. Oscar can finish the same puzzle in 3 hours working alone. How long would it take Oscar and Kenji to finish the puzzle if they worked on it together?
26. **Use Structure** A commercial jet flies 1,500 miles with the wind. In the same amount of time it can fly 1,000 miles against the wind. The speed of the jet in still air is 550 mph. Find the speed of the wind.



- Organize the given information and what you need to find in a table.
 - Write and solve a rational equation to find the wind speed.
27. **Reason** During their day at the beach, Jae and his friends rent a Jet Ski. They split the \$120 rental fee evenly among themselves. Then Jae, with only his friend Orion, share the cost of a \$16 pizza. If Jae spends a total of \$48 for both, then find the number of friends, n , with whom he shared the cost of the Jet Ski rental.
28. **Make Sense and Persevere** When driving to their family reunion, Rio's mom drove 10 miles at a rate of x mph and then 25 miles at a rate of $x + 10$ mph. The total driving time was 45 minutes. What were the two driving speeds at which Rio's mom drove?
29. **Generalize** So far this baseball season, Philip has gotten a hit 8 times out of 40 at-bats. He wants to increase his batting average to 0.333. Calculate the number of consecutive hits, h , he would need in order to achieve this goal. Round your answer to the nearest whole number.

ASSESSMENT PRACTICE

30. Which of the following rational equations have at least one extraneous solution? Select all that apply.

☐ $\frac{2}{x} = \frac{3}{x-4}$

☐ $\frac{x^2}{x-3} = \frac{9}{x-3}$

☐ $\frac{x-1}{x-5} = \frac{9}{x-5}$

☐ $x + \frac{3}{x} = 4$

☐ $\frac{x}{x-3} - \frac{3}{2} = \frac{3}{x-3}$

31. **SAT/ACT** Which of the following is the solution of $\frac{3}{x+1} = \frac{2}{x-3}$?

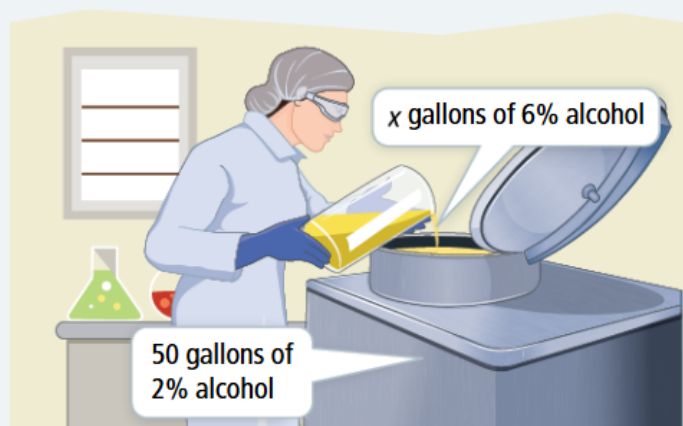
(A) $x = -11$

(B) $x = -\frac{7}{5}$

(C) $x = \frac{7}{5}$

(D) $x = 11$

32. **Performance Task** A chemist needs alcohol solution in the correct concentration for her experiment. She adds a 6% alcohol solution to 50 gallons of solution that is 2% alcohol. The function that represents the percent of alcohol in the resulting solution is $f(x) = \frac{50(0.02) + x(0.06)}{50 + x}$, where x is the amount of 6% solution added.



Part A How much 6% solution should be added to create a solution that is 5% alcohol?

Part B Use Appropriate Tools Explain the steps you could take to use your graphing calculator to verify the correctness of your answer to part (A).